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**Simulation of a Second-Order Phase Transition:
Paramagnet—Ferromagnet**

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Intoduction

First-order phase transitions have been known to mankind since prehistoric times. Over the millennia, we have ceased to be surprised by the grayish steam rising from a kettle or the icy crust covering rivers and canals in winter.

From this perspective, second-order phase transitions are significantly more complex. Due to the extremely low or high temperatures required for them to occur, not everyone gets to witness such a spectacle in person. The phenomena of superfluidity and superconductivity originate in the daunting realm of quantum physics.

Against their background, the second-order transition from paramagnet to ferromagnet stands apart; a fairly accurate model describing this process was proposed by Ernst Ising as early as 1924. In this work, we will attempt to analyze its basic principles and its solution using the Metropolis algorithm.

Main body

We can thank Michael Faraday for the modern classification of substances based on their susceptibility to an external magnetic field; in 1848, he identified three main categories of "magnetics."

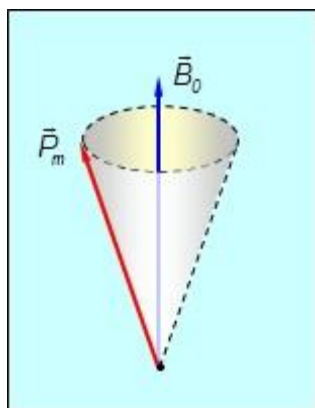


Figure 1

Diamagnets create their own field in response to an external one, directed in the opposite direction. The atoms of diamagnets do not possess an inherent magnetic moment, so there is nothing to orient along or against the field. Thus, an external magnetic field can only disturb the electron orbitals, causing them to precess (fig. 1). This additional movement of electrons is essentially an induced current that creates its own magnetic field directed against the external one. Many simple compounds, as well as almost all organic substances, are diamagnets. The phenomenon of diamagnetism itself is

inherent in almost all "magnetics," but it is the weakest possible interaction with an external magnetic field and remains unnoticed in para- and ferromagnets. Despite some interesting effects of diamagnetism, we will not consider it further.

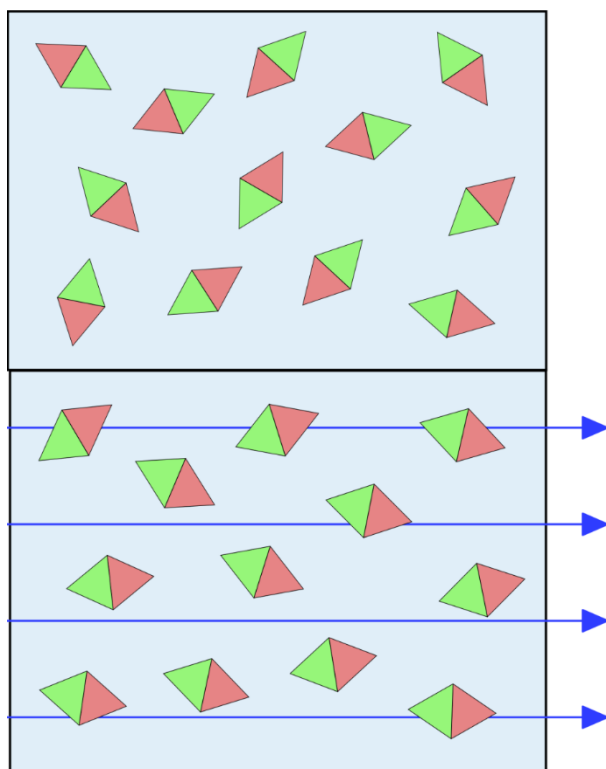


Figure 2

Unlike diamagnets, the atoms of paramagnetic substances already possess a certain non-zero magnetic moment. However, due to thermal motion, these moments are directed randomly, and as a result, at the macroscopic level, the sample does not exhibit significant susceptibility. Under the influence of an external field, the atomic moments partially align along the external field strength vector (fig. 2), but thermal motion prevents the magnetic "potential" (in a literary rather than physical sense) from being fully realized. Due to that same thermal motion, the moment the external field is turned off, the entire magnetization of the sample drops to zero.

Finally, ferromagnets are substances characterized by spontaneous magnetization—they can possess magnetic properties in the absence of an external field. Ferromagnets possess many unique properties, but the definition provided will be sufficient for our work.

The Transition

It follows from the above that the strength of the paramagnetic effect depends on the temperature and the strength of the external field. Under normal conditions, temperature and the external magnetic field compete for the material's magnetization. But is it possible to turn a paramagnet into a ferromagnet—to make the sample retain all its magnetic properties even without an external field?

As Pierre Curie discovered in 1895, such a transition indeed takes place at very low temperatures. The ordered arrangement of magnetic moments corresponds to a minimum of exchange energy (a purely quantum phenomenon, the nature of which we will not delve into further), and it is this striving for an energy minimum that can quiet the chaotic thermal motion at such temperatures.

To describe these processes, the Ising model utilizes the apparatus of statistical physics. The system we will work with can be represented as a lattice of particles, the magnetic moment of each being either 1 or -1. It is important to understand that temperature in this model is considered a constant; the system is in constant equilibrium with its environment. For simulation, we will have to consider the system in many different states, constantly making transitions between them. It is logical to record the probability of the system being in each state as the starting point of our reasoning (derived from the Boltzmann distribution)

$$p_u = \frac{1}{Z} e^{-\beta E_u},$$

where β — inverse of temperature, E_u — state's energy, and Z stands for the partition function:

$$Z = \sum_j e^{-\beta E_j},$$

Where j runs through each possible system state.

The partition function is an extremely useful, if not fundamental, quantity in statistical physics. However, for fans of numerical methods, it represents a "pain in the neck": even for a lattice of 10x10 molecules, a computer would have to calculate 2^{100} states and their energies, making the task impossible for any level of computing technology. Let's see how we can omit these calculations.

The system is always in equilibrium, which can be translated into the language of formulas: the probability of transitioning from one state to another is equal to the probability of the reverse transition:

$$p_u P(u \rightarrow v) = p_v P(v \rightarrow u),$$

What can be rewritten as follows:

$$\frac{P(u \rightarrow v)}{P(v \rightarrow u)} = \frac{p_v}{p_u} = e^{-\beta(E_v - E_u)}.$$

With relief, we note that during the transformations, the unpleasant partition function cancels out, and now all we need to calculate is the energy difference. At this step, we can proceed to the direct solution of the model.

Metropolis Algorithm

Each step of the simulation will attempt to "flip" one of the magnetic moments, and then, based on the derived patterns, move to a new state. The exchange energy of the system in a given state

$$E = - \sum_{\langle i,j \rangle} J s_i s_j$$

is calculated based as the sum each element's nearest neighbors — the energy associated with each individual element of the system is the sum of the product of its moment and the moments of its neighbors above, below, right, and left. J coefficient is here only for proportional purposes and in our simulations, we will equal it to zero.

Moving on to the most important part: the calculation of probabilities. Without loss of generality, let's consider two cases.

1. $E_v - E_u < 0$ — The energy of the new state is lower, making it more favorable. In this case, we transition to the new state without hesitation.
2. $E_v - E_u > 0$ — The transition is unfavorable from the perspective of minimizing exchange energy; however, there is still thermal motion hidden in the Boltzmann distribution. The probability of the transition is calculated accordingly. Notice that in this case the inverse transition is energetically beneficial, so $P(v \rightarrow u) = 1$, which gives balance equation:

$$\frac{P(u \rightarrow v)}{1} = e^{-\beta(E_v - E_u)},$$

$P(u \rightarrow v) = e^{-\beta(E_v - E_u)}$ which is a formula for our transition probability.

Program Implementation

Finally, after detailed development of the algorithm, it can be rewritten in program code. For ease of visualization, we choose the Processing environment, which uses the Java language. The lattice itself, with the necessary functions, will be written in a separate class. Its field would be a two-dimensional array `es` — the lattice itself, its dimensions `sx` and `sy`, floating point temperature `T` and `J` coefficient.

```
class Ising{
    private byte[][] es;
    private int sx, sy;
    private float T;
    private float J = 1f;
}
```

After a constructor, which doesn't require much attention here, we need to implement the Metropolis algorithm itself. To model transitions as honestly as possible, the "flipped" moment will be chosen randomly.

```
public void metropolis(){
    int ind_i = (int)random(0,sx);
    int ind_j = (int)random(0,sy);
    float dE = 0;
```

Also, note that to find the probability ratio, we only care about the change in energy. Knowing that these states differ by only one moment, the task can be reduced to calculating the energy associated only with that specific lattice element.

```
    if(ind_i>0)
        dE += J*(es[ind_i][ind_j]*es[ind_i-1][ind_j]); //left
    if(ind_i<sx-1)
        dE += J*(es[ind_i][ind_j]*es[ind_i+1][ind_j]); //right
    if(ind_j>0)
        dE += J*(es[ind_i][ind_j]*es[ind_i][ind_j-1]); //down
    if(ind_j<sy-1)
        dE += J*(es[ind_i][ind_j]*es[ind_i][ind_j+1]); //up

    if(dE<0)
        es[ind_i][ind_j]*=-1;
    else{
        if(random(0,1)<exp(-2*dE/T)){
            es[ind_i][ind_j]*=-1;
        }
    }
}
```

After a fairly simple visualization, we will also add magnetization to the output in order to see some numerical representation of the transition going. In physics, magnetization is calculated as the magnetic moment per unit volume; in our model, this can be calculated as the average moment of all elements in the:

```
float get_magnetisation(){
    float m = 0;
    for(int i = 0; i < size_x; i++)
        for(int j = 0; j < size_y; j++)
            m += es[i][j];
    return m / (size_x * size_y);
}
```

Regarding how many steps to take for visible results, there is no single answer. Intuitively, we will perform as many transitions "behind the scenes" per frame as there are constituent elements in our lattice. Thus, on average, during one frame, each element will try to flip, and the Metropolis algorithm will decide whether it should do so or not.:

```
void setup(){
    grid = new Ising(size_x,size_y,(byte)0,T,1);
}

void draw(){
    background(0);
    for(int i=0;i<size_x*size_y;i++)
        grid.metropolis();
    grid.render();
}
```


Results and Discussion

Let's see what we got: a 250 by 250 lattice with the initial orientation of magnetic moments chosen randomly. We start with an initial temperature of 2.52 K and observe complete chaos (fig. 3)

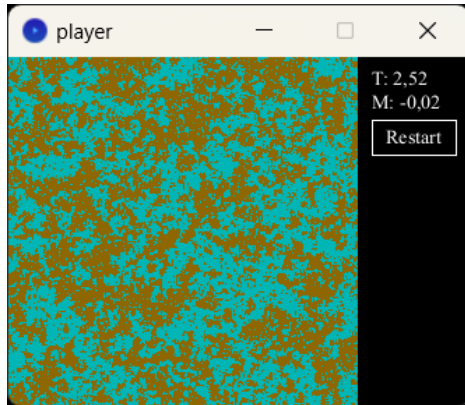
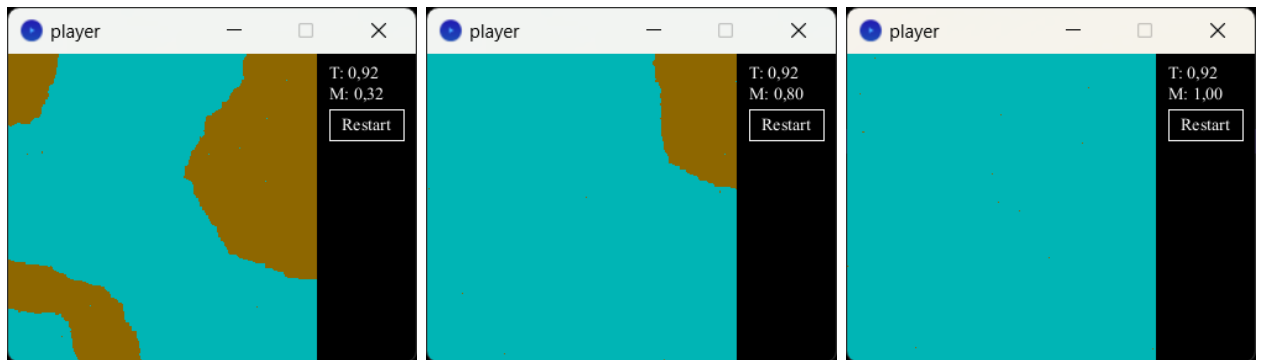


Figure 3

We abruptly drop the temperature to just below one Kelvin and see a more static picture. Magnetization has risen significantly. We step away for a few minutes and notice that it continues to strive toward one and reaches it (fig. 4 – 6). As expected, at such low temperatures, our initially paramagnetic material acquired spontaneous magnetization without the influence of an external field.



Figures 4 – 5 – 6 (from left to right)

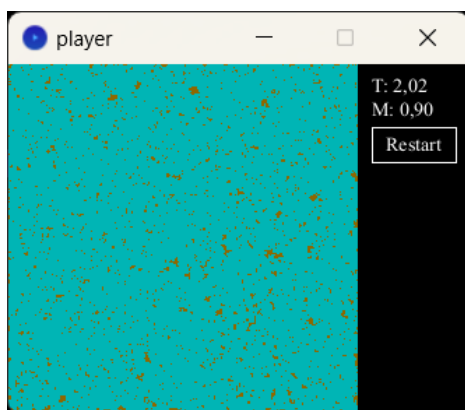


Figure 7

Let's try to destroy it again: we will gradually raise the temperature, hoping to notice the same transition but in the opposite direction. A temperature of 2.02 K does not yet yield results: despite the constant appearance of "spots" with opposite magnetic moment direction, magnetization remains near one (fig. 7)

However, a few more tenths of a Kelvin and the former order collapses in seconds (fig. 8)

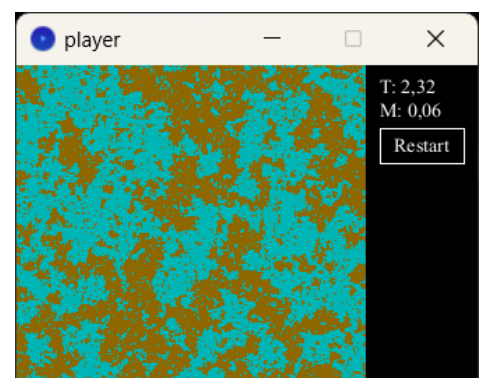


Figure 8

As we see on Figure 8, magnetization still maintains a positive value, but if we lower the temperature again, we see a completely different picture. Now the model tends toward aligning moments in the opposite direction (fig. 9-10)

During a second-order transition, the symmetry of the substance changes; after passing the Curie point, none of the preceding states can have an effect on the new one.



Figure 9

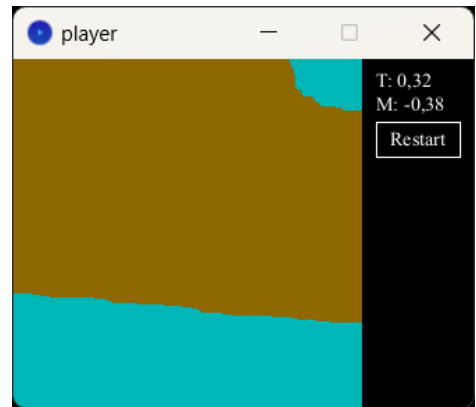


Figure 10

Code base

The original code of the research with some instructions on usage is put on GitHub here: https://github.com/serge-poliakoff/Ising_model

Conclusion

In our time, the phenomena studied by physicists often occur under difficult-to-achieve conditions or proceed too quickly or too slowly for experiments. In such cases, the use of computer models is essentially necessary: with their help, we can observe what is happening in more detail, put forward hypotheses, and draw parallels with already studied processes. The model developed by Ernst Ising in 1924 is one of the easiest to understand and implement in this regard, yet even today, some of its variations continue to baffle researchers.

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